Nonlinear Optimal Control Beyond the Hamilton–Jacobi–Bellman Equation

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***Abstract*—Within the framework of the linear quadratic regulator, it is well known that *"all roads lead to the Algebraic Riccati Equation."* The deceptive veil of *linearity* is torn herein by observing that such an algebraic equation generalizes to three different conditions in the nonlinear setting. The first one is obviously the celebrated Hamilton–Jacobi–Bellman partial differential equation arising by relying on dynamic programming arguments. However, it is shown that the optimal solution can be equivalently constructed also by characterizing a specific *invariant manifold* or an *invariant distribution* of the associated Hamiltonian vector field. While these three strategies reduce to the algebraic Riccati equation in the linear case, they instead lead to distinct conditions in the nonlinear setting, with the latter two yielding quasi-linear and linear partial differential equations, respectively, in place of the quadratic Hamilton–Jacobi–Bellman equation.**

***Index Terms*—Algebraic/geometric methods, invariant manifold and distribution, nonlinear systems, optimal control.**

I. Introduction

WHILE the celebrated *Brachystochrone problem*—posed in the form of a mathematical challenge by Johann Bernoulli more than three hundreds years ago—is commonly regarded as the historical birth of infinite-dimensional optimization possibly in the presence of differential constraints, the modern theory of optimal control has its roots in more practical problems faced in the mid of the 20th century, see [1] for an engaging review of such an evolution. The authors in [2], [3], and [4] provide instead extensive and detailed accounts of the mathematical technicalities involved in optimal control problems for linear and nonlinear dynamical models.

Most of the currently available methods to address optimal control problems may be traced back to two main strategies, either dynamic programming, see [5], or Pontryagin's minimum principle, see [6]. The former attempts to determine feedback control policies by characterizing, or approximating, the underlying *value function*, namely the minimum achievable cost from any initial condition, see, e.g., [7], [8], [9]. The latter instead relies on the solution of certain ordinary differential equations with split boundary conditions.

Depending on the length of the underlying interval, optimal control problems may be categorized into finite-horizon and infinite-horizon problems, see [10], [11] for discussions about the infinite-horizon setting. The latter framework is rather appealing in practice, as it allows, provided certain mild assumptions hold, to naturally and effectively achieve simultaneously optimality criteria and stability properties of desired equilibrium points. For such a class of problems, the optimal control law is constructed by either computing the value function, typically by solving a nonlinear (quadratic), first-order partial differential equation, i.e., the celebrated Hamilton–Jacobi–Bellman equation [2, Ch. 5], or by identifying specific trajectories of the underlying Hamiltonian dynamics restricted to invariant manifolds. The connection between the value function and the stable/unstable invariant submanifold of the Hamiltonian system has been elegantly established in [12]. The strategies proposed in [13] aim at providing approximate solutions to a class of partial differential equations by relying on the knowledge of (exact or approximate) flows of the underlying Hamiltonian dynamics on the stable invariant submanifold. The intrinsic connection between the optimal value function and the stable submanifold of the Hamiltonian dynamics has also been studied in [14] for the optimal balancing of a pendulum on a moving cart. Such an interplay has been leveraged in [15] to envision a fixed-point characterization of optimal control laws for linear and nonlinear systems alike.

In the case of linear-quadratic optimal control problems, despite the above observation that several alternative strategies are available to address the task (based on dynamic programming or Pontryagin's minimum principle), it seems that the algebraic Riccati equation (ARE) is actually ubiquitous. As a matter of fact, by quoting Willems [16], *"One often gets the impression that this equation in fact constitutes the bottleneck of all of linear system theory."*

The main contribution of the article is twofold.

First, the linear-quadratic regulator (LQR) problem, and in particular the role played by the ARE, is thoroughly revisited. It is argued that the reason why such an equation indeed always arises in the linear-quadratic case is essentially motivated by the need for certain conditions to hold *for all* x in the state space. Therefore, while the process of removing the dependence

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on the state variable yields the ARE, it should be acknowledged that instead the conditions leading to it are profoundly different whether one relies on dynamic programming or Pontryagin's minimum principle. These in fact hinge upon the characterization of the value function (see Section II-A) or of certain invariant hyperplanes (see Section II-B), respectively. By further building on this intuition, it is possible to derive yet another interpretation of the role of the ARE, which is pivotal to ensure that a suitably defined change of coordinates for the Hamiltonian dynamics renders the transformed system block upper triangular (see Section II-C). As a natural consequence of the latter observation, it is shown that the optimal control may be equivalently constructed by introducing a block lower triangular change of coordinates, associated with a *dual*[1] ARE. This intuition suggests that duality may actually be related to a choice of basis for the Hamiltonian dynamics and allows to envision a machinery to robustly compute the optimal costate variable as the solution of an asymptotically stable linear system of the same dimension as the original state (see Section III).

Second, the insights gathered by revisiting the linear framework are employed in the nonlinear case. While, as said above, the three strategies reduce to the ARE for linear systems, it is shown in Section IV that instead these generalize to profoundly different objects in the presence of nonlinear dynamics. This allows to extend the tool-kit of computational methods beyond the standard Hamilton–Jacobi–Bellman partial differential. In fact, it is shown that the optimal feedback may be equivalently constructed by characterizing certain invariant manifolds (see Section IV-B) or invariant distributions (see Section IV-C), which are distinct objects in the nonlinear framework, unlike their linear counterparts. Interestingly, the first strategy leads to a system of n *quasi-linear* partial differential equations, while the latter to a system of $2n$ *linear* partial differential equations, in place of the classic scalar *quadratic* Hamilton–Jacobi–Bellman partial differential equation. Finally, it is shown that the interpretation of the notion of duality mentioned above in the linear setting possesses two natural extensions within the nonlinear case (see Section V). Preliminary versions of the results in Sections III, IV-B, and V-A have been presented in [17].

The rest of this article is organized as follows. The role of the ARE in the linear-quadratic setting is thoroughly revisited in Section II, which also contains preliminary results and discussions. The objective of Section III is to show that the notion of duality in the above setting may be equivalently interpreted in terms of the choice of basis for the underlying Hamiltonian dynamics. Sections IV and V provide nonlinear counterparts of the results of the two previous sections. Finally, a few concluding remarks are discussed in Section VI.

Notation: Given a symmetric matrix $M = M^\top$, $M \succ 0$ ($M \succeq 0$, resp.) means that M is positive definite (positive semidefinite, resp.). Moreover, $\|v\|_M^2$ describes the norm (seminorm, resp.) defined by $v^\top M v$. Given a square matrix $M \in \mathbb{R}^{n\times n}$, $\operatorname{spec}(M)$ denotes the set of its eigenvalues. To avoid cumbersome expression and whenever this does not create any confusion, the notation (x_1, x_2), with $x_i \in \mathbb{R}^{n_i}$ describes the stacked vector $[x_1^\top \; x_2^\top]^\top$.

[1]The term *dual* is used for historical reasons, but the results do not rely on any formal notion of duality.

II. Role of the Algebraic Riccati Equation in Linear Quadratic Regulator

The main purpose of this section consists in establishing three distinct interpretations of the reasons why the ARE arises while attempting to solve the infinite-horizon LQR problem. Toward this end, consider the specially structured setting of linear dynamics and quadratic cost functional, namely the setting arising in the so-called LQR problem. Thus, consider the linear, time-invariant, system described by the equations

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x(0) = x_0 \tag{1}$$

with $x_0 \in \mathbb{R}^n$, where $x : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ and $u : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ denote the state and the control input, respectively. The performance of the control law is evaluated by means of the cost functional $J_q : \mathrm{C}^0[0,\infty) \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ defined as

$$J_q(u(\cdot), x_0) = \frac{1}{2}\int_0^\infty \left(\|x(t)\|_Q^2 + \|u(t)\|_R^2\right) dt \tag{2}$$

with $Q = Q^\top \succeq 0$ and $R = R^\top \succ 0$. The following standard assumption is considered.

Assumption 1: The pairs (A, B) and (A, C), with C such that $C^\top C = Q$, are stabilizable and detectable, respectively. Moreover, B is full column rank, namely $\operatorname{rank} B = m$. ∘

Provided that the structural Assumption 1 holds, it is well established (see, e.g., [2, Sect. 6.2.2]) that the optimal control law $u^\star : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ is given by

$$u^\star(t) = -R^{-1}B^\top P_+ x^\star(t) =: K^\star x^\star(t) \tag{3}$$

where $P_+ = P_+^\top$ is the unique positive semidefinite solution of the ARE (see also [4, Corollary 13.8])

$$0 = \underbrace{A^\top P + PA + Q - PBR^{-1}B^\top P}_{=:\mathcal{R}(P)}. \tag{4}$$

Furthermore, the optimal process $x^\star$ obeys the dynamics

$$\dot{x}^\star = \underbrace{(A - BR^{-1}B^\top P_+)}_{=A+BK^\star} x^\star, \quad x^\star(0) = x_0. \tag{5}$$

Note that the detectability properties of Assumption 1 and the structure of the cost (2) imply that the optimal process $x^\star$ tends asymptotically (in fact, exponentially) to zero. As is well known, the optimal control law (3) solving the LQR problem can be derived by relying on a few different arguments, mostly related to ideas inspired by dynamic programming, see, e.g., [5], [10], or borrowed from Pontryagin's minimum principle and the calculus of variations, see, for instance, [6] and [11]. Nonetheless, despite the availability of such alternative strategies, it is undeniable that the ARE (4) is indeed ubiquitous and intimately related to the construction of the optimal control law for the LQR problem, regardless of whether one is searching for a control law in the form of a state feedback [as in (3)] or of an open-loop input (see below in Section II-B). It is then particularly insightful to explore the reasons behind this phenomenon. This task is accomplished in the three following subsections, which discuss

as many different interpretations of the role of the ARE, and hence of its solution P_+, toward the construction of the optimal control law (3).

A. Role of ARE in Constructing Value Functions

Strategies inspired by dynamic programming arguments are based on the construction of the so-called value function $V : \mathbb{R}^n \to \mathbb{R}$, which provides, for each initial condition $x_0 \in \mathbb{R}^n$, the minimal achievable value of the cost functional (2), namely

$$x_0 \mapsto V(x_0) = \min_u \left\{ \frac{1}{2} \int_0^\infty (\|x(t)\|_Q^2 + \|u(t)\|_R^2) dt \right\} . \quad (6)$$

According to Bellman's original intuition, see [5], the value function is computed, in the linear-quadratic framework, by determining a quadratic function

$$V(x) = \frac{1}{2} x^\top P x \quad (7)$$

with the matrix $P = P^\top \succeq 0$ to be determined such that the equation

$$\begin{aligned} 0 = & \min_u \left\{ \frac{1}{2} x^\top Q x + \frac{1}{2} u^\top R u + \frac{\partial V}{\partial x}(Ax + Bu) \right\} \\ = & \min_u \left\{ \frac{1}{2} x^\top Q x + \frac{1}{2} u^\top R u + x^\top P(Ax + Bu) \right\} \\ = & \frac{1}{2} x^\top \left(Q + PA + A^\top P - PBR^{-1}B^\top P \right) x \\ = & \frac{1}{2} x^\top \mathcal{R}(P) x \end{aligned} \quad (8)$$

holds for all $x \in \mathbb{R}^n$, where the minimum with respect to u of the right-hand side of the second line is attained at (3). Therefore, the last identity is verified for all $x \in \mathbb{R}^n$ if and only if $\mathcal{R}(P) = 0$, hence if and only if the ARE (4) is satisfied.

B. Role of ARE in Constructing Invariant Hyperplanes

According to Pontryagain's minimum principle [6], which in turn has its roots in the theory of the calculus of variations, necessary and sufficient (by virtue of convexity of the underlying problem) conditions for the solution of the LQR problem are determined by computing trajectories of the Hamiltonian dynamics

$$\begin{bmatrix} \dot{x} \\ \dot{\lambda} \end{bmatrix} = \underbrace{\begin{bmatrix} A & -BR^{-1}B^\top \\ -Q & -A^\top \end{bmatrix}}_{=:H} \begin{bmatrix} x \\ \lambda \end{bmatrix} \quad (9)$$

with the property that the state variable x satisfies the boundary condition $x(0) = x_0$ and converges to zero as the time tends to infinity, together with some costate variable $\lambda : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$, as formalized below.

Lemma 1: Consider the LQR problem described by (1) and (2). Suppose that Assumption 1 holds. Then the optimal control law is given by

$$u_{\text{ol}}^\star(t) = -R^{-1}B^\top \lambda(t) \quad (10)$$

where λ satisfies (9) with the boundary conditions $x(0) = x_0$ and $\lim_{t\to+\infty} \lambda(t) = 0$. ○

Proof: To begin with, by introducing the Hamiltonian function $\mathbf{H}_q : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ associated with (1) and (2), namely

$$\mathbf{H}_q(x, \lambda, u) = \frac{1}{2} \left(x^\top Q x + u^\top R u \right) + \lambda^\top (Ax + Bu) \quad (11)$$

and by specializing the conditions of [10, Thm. 2.3] to the linear quadratic setting, one has that the costate λ satisfies the (lower block) equation in (9), whereas the control law $u_{\text{ol}}^\star$ is obtained as the (unique) minimizer of $\mathbf{H}_q$ with respect to u for fixed x and λ. Note then that, by the stabilizability property of Assumption 1, the cost $J_q(u_{\text{ol}}^\star)$ is finite, since there exists at least one control law $u(t) = \bar{K}x(t)$, with $\text{spec}(A + B\bar{K}) \subset \mathbb{C}^-$, such that J_q is bounded. Therefore, by the structure of $u_{\text{ol}}^\star$ in (10), it follows that:

$$\frac{1}{2} \int_0^\infty \left(x(t)^\top C^\top C x(t) + \lambda(t)^\top B R^{-1} B^\top \lambda(t) \right) dt < \infty$$

which in turn implies that $\lim_{t\to+\infty} Cx(t) = 0$ and $\lim_{t\to+\infty} \bar{R}B^\top \lambda(t) = 0$, with $\bar{R} = R^{-(1/2)}$, $\bar{R} = \bar{R}^\top$, such that $\bar{R}\bar{R} = R^{-1}$. Consider now the (virtual) linear output

$$y_v = \begin{bmatrix} 0 & \bar{R}B^\top \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} =: C_v \begin{bmatrix} x \\ \lambda \end{bmatrix} \quad (12)$$

associated with the Hamiltonian dynamics (9). A straightforward application of the observability PBH test reveals that (9) is detectable from the output y_v, and hence $\lim_{t\to+\infty} x(t) = 0$ and $\lim_{t\to+\infty} \lambda(t) = 0$, thus showing the claim. In fact, this is obtained by recalling the structural properties in Assumption 1 (including detectability of $(-A^\top, B^\top)$ by duality), and by observing that the observability properties of the pair (H, C_v) are equivalent to those of the pair $(H + GC_v, C_v)$, with

$$G = \begin{bmatrix} B\bar{R} & 0 \\ 0 & C^\top \end{bmatrix} \quad (13)$$

and, furthermore, that

$$\begin{aligned} \text{rank} \begin{bmatrix} A - sI & 0 \\ 0 & -A^\top - sI \\ 0 & \bar{R}B^\top \\ C & 0 \end{bmatrix} = & \text{rank} \begin{bmatrix} A - sI \\ C \end{bmatrix} \\ & + \text{rank} \begin{bmatrix} -A^\top - sI \\ B^\top \end{bmatrix} \end{aligned} \quad (14)$$

for all $s \in \mathbb{C}$. □

Lemma 1 establishes, on one hand, that the standard transversality condition (that is, requiring the costate to be equal to zero at the terminal time in the absence of terminal constraints or penalty on the state variable) is inherited by the infinite-horizon case (1), (2) and, on the other hand, that the optimal process is computed via the characterization of convergent trajectories of the Hamiltonian dynamics (9) for a given initial condition $x(0) = x_0$. It turns out that the above properties can be enforced

by restricting the dynamics (9) to an *invariant hyperplane* of the form

$$\mathcal{V} := \{(x, \lambda) \in \mathbb{R}^n \times \mathbb{R}^n : \lambda - Px = 0\} \tag{15}$$

for some matrix P to be determined, provided that the restricted dynamics $\dot{x} = (Ax - BR^{-1}B^\top \lambda)|_{\lambda - Px = 0} = (A - BR^{-1}B^\top P)x$ admit an asymptotically stable equilibrium at the origin. Therefore, invariance of the hyperplane $\mathcal{V}$ along the trajectories of (9) can be enforced by imposing that the condition

$$\begin{aligned} 0 = &\ (\dot{\lambda} - P\dot{x})|_{\lambda = Px} \\ = &\ (-Qx - A^\top \lambda - P(Ax - BR^{-1}B^\top \lambda))|_{\lambda = Px} \\ = &\ -(Q + A^\top P + PA - PBR^{-1}B^\top P)x \\ = &\ -\mathcal{R}(P)x \end{aligned} \tag{16}$$

holds for all $x \in \mathbb{R}^n$. Interestingly, a comment identical to that concluding Section II-A also applies to the arguments of this section: the last identity is verified for all $x \in \mathbb{R}^n$ if and only if $\mathcal{R}(P) = 0$, hence if and only if the ARE (4) is satisfied. Asymptotic stability of the restricted dynamics follows instead from standard properties of the unique stabilizing solution of the ARE (4), see, for instance, [4, Corollary 13.8].

C. Role of ARE in Constructing Triangularizing Change of Coordinates

The structural properties required in Assumption 1 ensure that the Hamiltonian matrix $H \in \mathbb{R}^{2n \times 2n}$ defined in (9) does not have eigenvalues on the imaginary axis, hence its spectrum is split into n eigenvalues with positive real part and n with negative real part. Therefore, as entailed by the constructions in Section II-B, the computation of the optimal control law from the Hamiltonian dynamics (9) tantamounts to restricting the dynamics of the latter onto a subspace $\mathcal{V}$ on which it only exhibits a stable behavior. A related, although different, perspective on the problem can be gained by aiming at isolating the stable from the unstable trajectories of the Hamiltonian dynamics and then by forcing the unstable subsystem to remain at rest for all times. Instead of restricting the dynamics (9) to an invariant hyperplane, this can be achieved by defining instead a *change of coordinates* of the form

$$\begin{bmatrix} x \\ \zeta \end{bmatrix} = \begin{bmatrix} x \\ -Px + \lambda \end{bmatrix}. \tag{17}$$

In fact, the Hamiltonian dynamics (9) are transformed via (17) into the system

$$\begin{bmatrix} \dot{x} \\ \dot{\zeta} \end{bmatrix} = \left[\begin{array}{c:c} A - BR^{-1}B^\top P & -BR^{-1}B^\top \\ \hdashline -\mathcal{R}(P) & -A^\top + PBR^{-1}B^\top \end{array}\right] \begin{bmatrix} x \\ \zeta \end{bmatrix}. \tag{18}$$

Thus, selecting P in (17) as a solution of the ARE (4) renders the dynamics (18) block upper triangular. Furthermore, if P is chosen as the unique stabilizing solution of $\mathcal{R}(P)$ one has that the unstable modes of H are those associated with the state variable ζ in the coordinates defined by (17). As a consequence, the optimal process may then be computed by avoiding that such modes contribute to the evolution of the state x in (18), namely by setting $\zeta(0) = 0$. The latter, in fact, due to the upper triangular structure of (18) ensures that $\zeta(t) = 0$ for all $t \geq 0$.

The insight gained via the discussions in Sections II-A, II-B, and II-C is summarized and more comprehensively organized in the following remark.

Remark 1: Despite the presence of multiple strategies to solve the LQR problem, one has the impression that all such methods ultimately lead to the ARE (4). Nonetheless, while this intuition may appear valid if one focuses on the *last* step of each construction, namely that requiring the solution of the algebraic equation $\mathcal{R}(P) = 0$ – it is particularly instructive to inspect the *penultimate* step instead.

More precisely, in Section II-A one searches for a scalar function $V : \mathbb{R}^n \to \mathbb{R}$ and the penultimate step requires that another scalar function, quadratic in the state variable x, is identically equal to zero; in Section II-B one searches for a (n-dimensional) vector-valued mapping $h : \mathbb{R}^n \to \mathbb{R}^n$ ($h(x) = Px$ in the linear setting) and the penultimate step requires that a vector-valued mapping, linear in the state variable x, is identically equal to zero along trajectories of the Hamiltonian system (9); in Section II-C one searches for a ($2n$-dimensional) vector-valued mapping $\Phi : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ ($\Phi(x, \lambda) = (x, -Px + \lambda)$ in the linear setting) such that the Hamiltonian system becomes block upper triangular in the coordinates defined by Φ.

Once the state variable x is removed from the picture (since the conditions must hold for all $x \in \mathbb{R}^n$) all the requirements boil down to the same equation. However, before removing the dependence on the state variable x, the conditions actually possess profoundly different natures. These differences become crucial in the nonlinear setting, as extensively explained in Section IV. ▲

The implications and intuition behind the constructions discussed in Section II-C are further extended here, before the nonlinear case is dealt with.

III. Optimal Process via a Dual Riccati Equation

According to the third interpretation of the ARE (4) provided in Section II-C, it appears that the role of the matrix P consists in transforming the Hamiltonian dynamics into a block *upper triangular* system, see (18). It is then reasonable to wonder whether also the opposite design is viable, i.e., transforming (9) into a block *lower triangular* system, and on the structure of the resulting conditions. These questions are addressed in this section by characterizing the optimal process $x^\star(t)$, $t \in [0, \infty)$, via a dual ARE.

Proposition 1: Consider the LQR problem described by (1) and (2). Suppose that Assumption 1 holds with the pair (A, C) observable. The optimal process is given by

$$x^\star(t) = M_+ \lambda(t) \tag{19}$$

with $\lambda : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ governed by

$$\dot{\lambda}(t) = -(A + M_+ Q)^\top \lambda(t), \quad \lambda(0) = M_+^{-1} x_0 \tag{20}$$

where $M_+ = M_+^\top \in \mathbb{R}^{n \times n}$ is the unique positive definite solution of the *dual* ARE

$$0 = \underbrace{AM + MA^\top - BR^{-1}B^\top + MQM}_{=: \mathcal{N}(M)}. \tag{21}$$

○

Proof: To begin with, note that the set of nonsingular solutions of (21) coincides with the set of nonsingular solutions of the ARE (4), as it is immediately obtained by left and right multiplying both sides of the former by M^{-1}. The equivalence, in turn, implies that (21) admits a unique positive definite solution provided Assumption 1 is satisfied with the hypothesis on the pair (A, C) further strengthened to observability. Therefore, such a unique solution is given by the inverse of the unique positive definite solution of (4), namely $M_+ = P_+^{-1}$. Recall then that the optimal process $x^\star : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ can be obtained as the solution to the Hamiltonian dynamics

$$\begin{bmatrix} \dot{x}^\star \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} A & -BR^{-1}B^\top \\ -Q & -A^\top \end{bmatrix} \begin{bmatrix} x^\star \\ \lambda \end{bmatrix}, \quad \begin{bmatrix} x^\star(0) \\ \lambda(0) \end{bmatrix} = \begin{bmatrix} I \\ P_+ \end{bmatrix} x_0 \tag{22}$$

where $\lambda : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes the costate variable of the lifted dynamics (22), see also the discussion in Section II-B. By considering now the change of coordinates described by

$$\begin{bmatrix} \xi \\ \lambda \end{bmatrix} = \begin{bmatrix} x^\star - M\lambda \\ \lambda \end{bmatrix} = \underbrace{\begin{bmatrix} I & -M \\ 0 & I \end{bmatrix}}_{=:T} \begin{bmatrix} x^\star \\ \lambda \end{bmatrix} \tag{23}$$

it is straightforward to observe that the dynamics (22) becomes in the transformed coordinates

$$\begin{bmatrix} \dot{\xi} \\ \dot{\lambda} \end{bmatrix} = \left[\begin{array}{c:c} A + MQ & \mathcal{N}(M) \\ \hdashline -Q & -(A+MQ)^\top \end{array}\right] \begin{bmatrix} \xi \\ \lambda \end{bmatrix}. \tag{24}$$

It can then be noted that employing any solution of the dual ARE (21) in the construction of the change of coordinates (23) transforms the Hamiltonian dynamics into a block lower triangular system. Furthermore, adapting also the boundary conditions of (22) to the coordinates in (23), one has that $(\xi(0), \lambda(0)) = (0, M_+^{-1}x_0)$. The latter, together with the block lower triangular structure of (24), ensures that $\xi(t) = 0$ for all $t \in \mathbb{R}_{\geq 0}$. Therefore, by inspecting the first row of (23), it follows that $0 = \xi(t) = x^\star(t) - M_+\lambda(t)$ for all $t \in \mathbb{R}_{\geq 0}$, thus showing the claim. □

Remark 2: The proof of Proposition 1 reveals that the motivation behind the matrix equation (21) is to enforce a block *lower triangular* structure for the transformed Hamiltonian dynamics, so that only a component of the state/costate variable becomes relevant toward the computation of the optimal process. Complementing the discussion in Section II-C, the above reasoning suggests the intuition that the occurrence of the standard ARE or of the dual ARE is simply related to a choice of basis for the Hamiltonian dynamics (22). The above observation may be of interest in light of the fact that, historically, dual Riccati equations have typically been associated with the filtering problem, whereas (standard) Riccati equations are inextricably related to the LQR problem. ▲

While the statement of Proposition 1 suggests an abstract characterization of the optimal process, the main objective of the following result is to discuss how the knowledge of (19) and (20) can be leveraged to design the optimal control law, which recovers the synthesis suggested by arguments inspired by Pontryagin's minimum principle, namely (10) with λ given by (22), albeit with a crucial difference that is illustrated below (see Proposition 3). The next statement shows that the property ensuring that the variable ξ remains identically equal to zero for all times along the optimal process contains sufficient information to construct the underlying optimal control law.

Proposition 2: Consider the LQR problem described by (1) and (2). Suppose that Assumption 1 holds with the pair (A, C) observable. Then the optimal control law is given by

$$\begin{aligned} u^\star(t) &= -B^\dagger(AM_+\lambda(t) - M_+\dot{\lambda}(t)) \\ &= -R^{-1}B^\top\lambda(t) \end{aligned} \tag{25}$$

with $\lambda(t)$, $t \in [0, \infty)$ obtained via (20), where $B^\dagger \in \mathbb{R}^{m\times n}$ denotes the left pseudoinverse of B, i.e., $B^\dagger = (B^\top B)^{-1}B^\top$. ○

Proof: Let $\eta = x - M_+\lambda$, with x denoting the state variable of (1). By inspecting (19) it follows that the optimal design of u in (1) should be such that $\eta \equiv 0$, namely such that $x \equiv x^\star$. The latter identity is satisfied at $t = 0$ since $\eta(0) = x(0) - M_+\lambda(0) = x_0 - M_+M_+^{-1}x_0 = 0$. Therefore, consider the time derivative of η, namely

$$\begin{aligned} \dot{\eta} &= Ax + Bu - M_+\dot{\lambda} \\ &= A(\eta + M_+\lambda) + Bu - M_+\dot{\lambda} \\ &= A\eta + (AM_+\lambda - M_+\dot{\lambda}) + Bu\,. \end{aligned} \tag{26}$$

Since $M_+\lambda(t)$ coincides with the optimal process $x^\star(t)$ for all $t \geq 0$, by Proposition 1, it follows that also $M_+\dot{\lambda}(t) = \dot{x}^\star(t)$. Furthermore, by recalling (5), one has that the identity $\dot{x}^\star(t) - Ax^\star(t) = Bu^\star(t)$ clearly holds for all $t \geq 0$, with $u^\star$ denoting the optimal control law. As a consequence, since $x^\star(t) = M_+\lambda(t)$ and $\dot{x}^\star(t) = M_+\dot{\lambda}(t)$, the condition

$$Ax^\star(t) - \dot{x}^\star(t) = AM_+\lambda(t) - M_+\dot{\lambda}(t) \in \operatorname{Im} B \tag{27}$$

holds for all $t \in [0, \infty)$. Thus, the property (27) implies that enforcing $(AM_+\lambda - M_+\dot{\lambda}) + Bu \equiv 0$ (where the left-hand side contains terms appearing in the last line of (26)) consists of a compatible system of equations for all time $t \in \mathbb{R}_{\geq 0}$, which are solved by the first line of (25). Furthermore, by relying on the right-hand side of (20), the above control law may be written as

$$\begin{aligned} u^\star &= -B^\dagger(AM_+ + M_+A^\top + M_+QM_+)\lambda \\ &= -B^\dagger(BR^{-1}B^\top)\lambda = -R^{-1}B^\top\lambda \end{aligned} \tag{28}$$

where the first identity is obtained by virtue of (21). □

Despite the evident similarity between (25) and the standard (open-loop) control law yielded by the Pontryagin's minimum principle (10), it is worth observing that the former is, in fact, obtained via forward propagation of the *restricted*, i.e., of dimension n, dynamics (20) in place of the entire Hamiltonian dynamics (22). Such a difference becomes crucial when recalling that the latter dynamics are intrinsically unstable, whenever Assumption 1 holds. The following statement instead discusses the stability properties of the zero equilibrium of the dynamics (20) that governs the evolution of the optimal costate.

Proposition 3: Consider the system (20) and suppose that Assumption 1 holds with the pair (A, C) observable. Then $\lambda =$

0 is a globally exponentially stable equilibrium of the system (20). ∘

Proof: The claim is established by showing that the eigenvalues of the matrix $-(A+M_+Q)^\top$ belong to the open left half-plane of $\mathbb{C}$. In fact, the spectrum of $-(A+M_+Q)^\top$ is such that

$$\begin{aligned}\operatorname{spec}(-(A+M_+Q)^\top) &= \operatorname{spec}(-(A^\top+QM_+))\\ &= \operatorname{spec}(-M_+(A^\top+QM_+)M_+^{-1})\\ &= \operatorname{spec}((AM_+-BR^{-1}B^\top))M_+^{-1})\\ &= \operatorname{spec}(A-BR^{-1}B^\top P_+) \qquad (29)\end{aligned}$$

where the dual ARE (21) is employed in the third identity and the last identity holds by virtue of the fact that $M=P_+^{-1}$. By standard properties of the solution P_+ of (4) it follows that $\operatorname{spec}(-(A+M_+Q)^\top)\subset\mathbb{C}^-$. □

A few comments about the results above are in order.

Remark 3: In light of the conclusions of Propositions 1 and 3, the dynamical system (20) can be interpreted and employed as a *robust* mechanism to generate the optimal costate. Indeed, any numerical error in the computation of the matrix M_+ or in the selection of the initial condition $\lambda(0)=M_+^{-1}x_0$ is compensated by the asymptotic stability properties of (20). This permits the open-loop *direct* computation of the optimal costate in a robust manner, instead of resorting to the identity $\lambda(t)=P_+e^{(A+BK^\star)t}x_0$—derived by virtue of the structure of (15) and the definition of (5)—yielding the robust open-loop control $u(t)=-R^{-1}B^\top P_+e^{(A+BK^\star)t}x_0$.

Furthermore, the definition of the system (20) implies that the time history of the optimal costate is given by $\lambda(t)=e^{-(A+M_+Q)^\top t}M_+^{-1}x_0$. Thus, since $x^\star(t)=e^{(A+BK^\star)t}x_0$ by (5), it follows that:

$$\lambda(t)=e^{-(A+M_+Q)^\top t}M_+^{-1}e^{-(A+BK^\star)t}x^\star(t)\,. \qquad (30)$$

By putting (30) into perspective with respect to the definition of the invariant hyperplane $\mathcal{V}$ in (15), one has that the identity

$$P_+=e^{-(A+M_+Q)^\top t}M_+^{-1}e^{-(A+BK^\star)t} \qquad (31)$$

holds for all $t\geq 0$. ▲

IV. Optimal Control of Nonlinear Systems

In this section, the results presented in Sections II and III, within the setting of linear systems, are extended to nonlinear systems. Toward this end, consider an input-affine nonlinear system described by the equations

$$\dot{x}(t)=f(x(t))+g(x(t))u(t),\qquad x(0)=x_0 \qquad (32)$$

where $x:\mathbb{R}_{\geq 0}\to\mathbb{R}^n$ and $u:\mathbb{R}_{\geq 0}\to\mathbb{R}^m$ denote the state and input, respectively; with $f:\mathbb{R}^n\to\mathbb{R}^n$, $f(0)=0$, and $g:\mathbb{R}^n\to\mathbb{R}^{n\times m}$. The system (32) is associated with the cost functional $J:\mathrm{C}^0[0,\infty)\times\mathbb{R}^n\to\mathbb{R}_{\geq 0}$ defined as

$$J(u(\cdot),x_0)=\int_0^\infty\left(q(x(t))+\frac{1}{2}\|u(t)\|^2\right)dt \qquad (33)$$

where $q:\mathbb{R}^n\to\mathbb{R}_{\geq 0}$. With respect to a more general formulation of the cost, here the weighting matrix of the penalty of the control input is selected as the identity matrix (namely $R=I$) to ease the notation of the subsequent derivations: this can be immediately generalized with the appropriate, straightforward, adaptations of the results. The following assumption ensures, together with the structure of (32), that the LQ approximation of (32) and (33) admits a unique optimal solution.

Assumption 2: The vector field f, the mapping g and the function q are sufficiently smooth. The pair $(\bar{A},\bar{B})$, with

$$\bar{A}:=\frac{\partial f}{\partial x}(x)\Big|_{x=0},\quad \bar{B}:=g(x)\Big|_{x=0} \qquad (34)$$

is stabilizable and the pair $(\bar{A},\bar{Q})$, with $\bar{A}$ in (34) and

$$\bar{Q}:=\frac{\partial^2 q}{\partial x^2}(x)\Big|_{x=0} \qquad (35)$$

is detectable. ∘

The optimal control law is characterized in the following three subsections, which mimic the organization of Sections II-A, II-B, and II-C, respectively. While the three distinct paths all lead to the ARE (4), i.e., to $\mathcal{R}(P)=0$, in the case of linear systems, the discussions below show that instead the straightforward extensions of the constructions in each subsection above provide profoundly different conditions in the nonlinear setting. In particular, while the constructions recalled in Section IV-A yield the well-known Hamilton–Jacobi–Bellman partial differential equation, the results presented in Sections IV-B and IV-C establish new and deeper connections among several tools and ingredients arising in nonlinear control, such as invariant manifold, invariant distributions, and their computation.

A. Nonlinear Optimal Control via Value Function

The most natural extension of the arguments employed in Section II-A consists in resorting to the characterization of the underlying *value function* $V^\star:\mathbb{R}^n\to\mathbb{R}$, defined as

$$x_0\mapsto V^\star(x_0)=\min_u J(u(\cdot),x_0) \qquad (36)$$

with J given in (33). Along the lines of the regularity requirements introduced by Assumption 2, suppose that also the following property is verified.

Assumption 3: The value function $V^\star$ is twice continuously differentiable for all $x\in\mathbb{R}^n$. ∘

Note that, in light of the regularity properties implied by Assumption 2, the condition of Assumption 3 indeed holds at least in a neighborhood of the origin, see, e.g., [7]. The requirement of Assumption 3 ensures that the value function may be effectively characterized by solving the corresponding Bellman's equation of dynamic programming that, in the setting of (32) and (33), becomes (compare with the equation in the first line of (8))

$$0=\min_u\left\{q(x)+\frac{1}{2}u^\top u+\frac{\partial V^\star}{\partial x}(x)(f(x)+g(x)u)\right\} \qquad (37)$$

for all $x\in\mathbb{R}^n$, with $V^\star(0)=0$ together with the need for determining a positive semidefinite solution of (37). By explicitly performing the (unconstrained) minimization task indicated in

the right-hand side of (37), the latter equation reduces to the celebrated Hamilton–Jacobi–Bellman PDE

$$0 = q(x) + \frac{\partial V^\star}{\partial x}(x) f(x) - \frac{1}{2}\frac{\partial V^\star}{\partial x}(x) g(x) g(x)^\top \frac{\partial V^\star}{\partial x}(x)^\top \tag{38}$$

for all $x \in \mathbb{R}^n$ (or for all x in a suitably defined neighborhood of the origin). Moreover, the optimal process $x^\star$ is then governed by the autonomous dynamics

$$\dot{x}^\star = \underbrace{f(x^\star) - g(x^\star) g(x^\star)^\top \frac{\partial V^\star}{\partial x}(x^\star)^\top}_{=:\, f_o(x^\star)}, \quad x^\star(0) = x_0\,. \tag{39}$$

Standard asymptotic stability properties of the optimal closed loop (39) are implied by the following assumption.

Assumption 4: The system (39) is zero-state detectable via the output $y_v = q(x)$, i.e., $y_v \equiv 0$ implies $\lim_{t\to+\infty} x(t) = 0$. ∘

We conclude with the key observation.

> The specific occurrence of the ARE in Section II-A generalizes in the nonlinear setting to a *single, quadratic, partial differential equation in the unknown function* $V^\star : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$.

B. Nonlinear Optimal Control via Invariant Manifold

With the objective of replicating the constructions of Section II-B, related to the computation of a certain invariant hyperplane, note that the (minimized) Hamiltonian function $\mathbf{H} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ associated with (32) and (33) is given by

$$\mathbf{H}(x, \lambda) = q(x) + \lambda^\top f(x) - \frac{1}{2}\lambda^\top g(x) g(x)^\top \lambda\,. \tag{40}$$

The function $\mathbf{H}$ is further naturally associated with the canonical Hamiltonian dynamics

$$\Sigma_{\mathbf{H}} : \begin{cases} \dot{x} = f(x) - g(x) g(x)^\top \lambda \\ \dot{\lambda} = -\frac{\partial q}{\partial x}(x)^\top - \frac{\partial f}{\partial x}(x)^\top \lambda + \sum_{i=1}^m \lambda^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \lambda \end{cases} \tag{41}$$

where $g_i : \mathbb{R}^n \to \mathbb{R}^n$ denotes the ith column of the matrix-valued function g. The optimal process $(x^\star(t), \lambda(t))$ for all $t \in \mathbb{R}_{\geq 0}$ is characterized, in light of Assumption 3, by the Hamiltonian dynamics (41) together with certain (initial) boundary conditions that are related to the gradient of the value function, namely

$$\begin{cases} x(0) = x_0 \\ \lambda(0) = \frac{\partial V^\star}{\partial x}(x_0)^\top. \end{cases} \tag{42}$$

Furthermore, by the regularity hypotheses of Assumption 3 it follows that the costate variable λ coincides for any time $t \in \mathbb{R}_{\geq 0}$ with the gradient of the value function $V^\star$ with respect to x along the optimal trajectory. Note now that positive semidefiniteness of the function $V^\star$ together with the condition $V^\star(0) = 0$ ensures that also the gradient of $V^\star$ with respect to x vanishes at the origin. Therefore, provided Assumptions 2–4 are verified, the optimal costate satisfies $\lim_{t\to+\infty} \lambda(t) = 0$.

The above discussion reveals that the solution to the nonlinear optimal control problem described by (32) and (33) revolves around the characterization of a specific (convergent) trajectory of the Hamiltonian dynamics (41), belonging to the stable invariant manifold of (41). Thus, by mimicking and extending the constructions of Section II-B in the linear setting, one could attempt to isolate such a specific trajectory by directly characterizing the stable invariant manifold, described by [compare also with the definition of $\mathcal{V}$ in (15) for linear systems]

$$\mathcal{M} := \{(x, \lambda) \in \mathbb{R}^n \times \mathbb{R}^n : \lambda - h(x) = 0\} \tag{43}$$

with the mapping $h : \mathbb{R}^n \to \mathbb{R}^n$ to be defined, as discussed in the following statement.

Proposition 4: Consider the optimal control problem described by (32) and (33), and suppose that Assumptions 2–4 hold. Suppose that the mapping $h : \mathbb{R}^n \to \mathbb{R}^n$, $h \in \mathcal{C}^1$, $h(0) = 0$, satisfies the quasi-linear PDE

$$\begin{aligned} 0 = {} & \frac{\partial h}{\partial x}(x)(f(x) - g(x) g(x)^\top h(x)) + \frac{\partial q}{\partial x}(x)^\top \\ & + \left(\frac{\partial f}{\partial x}(x)^\top - \sum_{i=1}^m h(x)^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \right) h(x) \end{aligned} \tag{44}$$

for all $x \in \mathbb{R}^n$, together with the boundary condition

$$\frac{\partial h}{\partial x}(0) = P_+\,. \tag{45}$$

Then there exists a nonempty set $\Omega \subset \mathbb{R}^n$, containing the origin, such that the optimal process $x^\star(t)$, $t \in \mathbb{R}_{\geq 0}$, is governed by

$$\dot{x}^\star(t) = f(x^\star(t)) - g(x^\star(t)) g(x^\star(t))^\top h(x^\star(t)), \quad x^\star(0) = x_0 \tag{46}$$

for all $x_0 \in \Omega$, and hence $u^\star(x) = -g(x)^\top h(x)$. ∘

Proof: By Assumptions 2 and 3 the optimal process $(x^\star(t), \lambda(t))$, for $t \in \mathbb{R}_{\geq 0}$ and for all initial conditions x_0 in a neighborhood of the origin, is uniquely characterized by the trajectory of (41) passing through (42) at the initial time. Consider the variable $\zeta = \lambda - h(x)$ suggested by the structure of $\mathcal{M}$ in (43). In light of (41), the dynamics of ζ are

$$\begin{aligned} \dot{\zeta} = {} & \left(\dot{\lambda} - \frac{\partial h}{\partial x}(x) \dot{x} \right) \Big|_{\lambda = \zeta + h(x)} \\ = {} & \left(-\frac{\partial q}{\partial x}(x)^\top - \frac{\partial f}{\partial x}(x)^\top \lambda + \sum_{i=1}^m \lambda^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \lambda \right. \\ & \left. - \frac{\partial h}{\partial x}(x) \left(f(x) - g(x) g(x)^\top \lambda \right) \right) \Big|_{\lambda = \zeta + h(x)} \\ = {} & -\frac{\partial f}{\partial x}(x)^\top \zeta + \sum_{i=1}^m \zeta^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \zeta + \frac{\partial h}{\partial x}(x) g(x) g(x)^\top \zeta \\ & + \sum_{i=1}^m h(x)^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \zeta + \sum_{i=1}^m \zeta^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top h(x) \\ =: {} & \delta(x, \zeta) \end{aligned} \tag{47}$$

with the continuous mapping $\delta : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ having the property that $\delta(x, 0) = 0$ for all $x \in \mathbb{R}^n$, where the third equality

is obtained by virtue of (44). The structure of the dynamics governing the auxiliary variable ζ in (47) ensures that the selection of the initial condition according to $\zeta(0) = 0$ implies that $\zeta \equiv 0$, namely $\zeta(t)$ is equal to zero for all $t \in \mathbb{R}_{\geq 0}$ along the trajectories of (41). As a consequence, the set $\{(x, \zeta) \in \mathbb{R}^n \times \mathbb{R}^n : 0 = \zeta\}$, or equivalently $\{(x, \lambda) \in \mathbb{R}^n \times \mathbb{R}^n : 0 = \lambda - h(x)\}$, with the mapping h solving (44), is an invariant manifold for the dynamics (41). Furthermore, the restriction of $\Sigma_{\mathbf{H}}$ to the above invariant manifold is described by the dynamics $\dot{x} = f(x) - g(x)g(x)^\top h(x)$, which possess a locally exponentially stable equilibrium at the origin. In fact, by (45), the linear terms of the vector field $f(x) - g(x)g(x)^\top h(x)$ are $(A - BB^\top P_+)x$, with the matrix $(A - BB^\top P_+)$ being Hurwitz due to the properties of P_+ implied by Assumption 2. Finally, by the boundary condition (45), the manifold is tangent at the origin to the stable subspace of the linearization of (41), which is a well-defined n-dimensional subspace in light of Assumption 2, hence locally coincides with the stable submanifold of (41) by uniqueness of the latter. Therefore, the proof is concluded by observing that for all x_0 in a neighborhood of the origin, the (initial) state $(x_0, h(x_0))$ belongs to the stable invariant submanifold, and hence the optimal process satisfies (46) together with the further constraint $\zeta \equiv 0$. □

Remark 4: Consider the value function $V^\star$ associated with the nonlinear optimal control problem described by (32) and (33), and suppose that Assumption 2 holds. Within this framework, the value function $V^\star$ is then provided by the Hamilton–Jacobi–Bellman equation (38). Since the right-hand side of the latter equation (38) must be identically equal to zero for all x, or in a certain set, then within the same set also its partial derivative with respect to x should be equal to zero. Thus, considering the partial derivatives of both sides of (38), one obtains

$$\begin{aligned} 0 = \; & \frac{\partial q}{\partial x}(x)^\top + \frac{\partial^2 V^\star}{\partial x^2}(x) f(x) + \frac{\partial f}{\partial x}(x)^\top \frac{\partial V^\star}{\partial x}(x)^\top \\ & - \sum_{i=1}^{m} \frac{\partial V^\star}{\partial x}(x) g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \frac{\partial V^\star}{\partial x}(x)^\top \\ & - \frac{\partial^2 V^\star}{\partial x^2}(x) g(x) g(x)^\top \frac{\partial V^\star}{\partial x}(x)^\top \end{aligned} \tag{48}$$

where the existence of the second derivative of $V^\star$ with respect to x is guaranteed by Assumption 3. The comparison between (48) and (44) then reveals that the gradient of the value function $V^\star$ is indeed a solution to (44) that additionally satisfies the boundary condition (45). ▲

Remark 5: By employing the state variable x and the *off-the-manifold* variable ζ to express the dynamics (41), a straightforward adaption of the reasoning in the proof of Proposition 4 shows that $\Sigma_{\mathbf{H}}$ becomes

$$\begin{cases} \dot{x} = f(x) - g(x)g(x)^\top (h(x) + \zeta) \\ \dot{\zeta} = \delta(x, \zeta). \end{cases} \tag{49}$$

A straightforward inspection of (49) suggests an interesting difference with respect to (18), which is obtained via a conceptually similar change of coordinates (17) although in the linear setting. While the latter in fact exhibit a block-triangular structure, the former system (49) is instead not such that the dynamics of ζ are *decoupled* from the state variable x, hence the resulting nonlinear system is not triangular. Therefore, expanding the discussion at the end of Remark 2, it is worth observing that the conditions in Proposition 4 for the nonlinear case revolve around the characterization of an invariant *manifold* rather than an invariant *distribution:* such a distinction cannot be appreciated in the linear setting ▲

We conclude with the key observation.

> The specific occurrence of the ARE of Section II-B generalizes in the nonlinear setting to a *system of n, quasi-linear partial differential equations in the unknown mapping* $h : \mathbb{R}^n \to \mathbb{R}^n$.

Example 1: To illustrate the constructions discussed in this section consider the scalar nonlinear system

$$\dot{x} = x^2 + u \tag{50}$$

with $x : \mathbb{R} \to \mathbb{R}$ and $u : \mathbb{R} \to \mathbb{R}$, namely as in (32) with $f(x) = x^2$ and $g(x) = 1$, together with the cost functional (33) with $q(x) = (1/2)x^2$. The main motivation for considering such an optimal control problem consists in the fact that it admits a closed-form expressions for the optimal value function and feedback control law. More precisely

$$V^\star(x) = \frac{1}{3}(x^2 + 1)^{(3/2)} + \frac{1}{3}x^3 - \frac{1}{3} \tag{51}$$

for all $x \in \mathbb{R}$. The linear quadratic approximation of the optimal control problem is such that the corresponding ARE (4) yields $P_+ = 1$. The quasi-linear partial differential equation (44) is given by

$$0 = \frac{dh}{dx}(x)(x^2 - h(x)) + x(1 + 2h(x)), \; \frac{dh}{dx}(0) = 1 \tag{52}$$

which is indeed satisfied by the derivative of $V^\star$ in (51). Let now $h^{(\kappa)} : \mathbb{R} \to \mathbb{R}$ denote a function with the property that the right-hand side of (52) with h replaced by $h^{(\kappa)}$ becomes $o(|x|^{\kappa+1})$. Fig. 1 depicts the derivative of the optimal value function $V^\star$ (solid black line) together with $h^{(\kappa)}$, for $\kappa = 1, 2, 3, 5$ (since $h^{(4)} \equiv h^{(3)}$). In particular, it is straightforward to see that

$$h^{(5)}(x) = x + x^2 + \frac{1}{2}x^3 - \frac{1}{8}x^5 \tag{53}$$

solves (52) up to terms of order five and yields an accurate approximation of the gradient of the value function. △

Example 2: Consider the nonlinear optimal control problem involving the Van der Pol oscillator (32) with

$$f(x) = \begin{bmatrix} x_2 \\ -x_1 - \mu(1 - x_1^2)x_2 \end{bmatrix}, \quad g(x) = \begin{bmatrix} 0 \\ x_1 \end{bmatrix} \tag{54}$$

with $\mu = 1/2$, and the cost functional (33) with $q(x) = (1/2)x_2^2$. The linearized model is described by

$$\bar{A} = \begin{bmatrix} 0 & 1 \\ -1 & -\frac{1}{2} \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \bar{Q} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \tag{55}$$

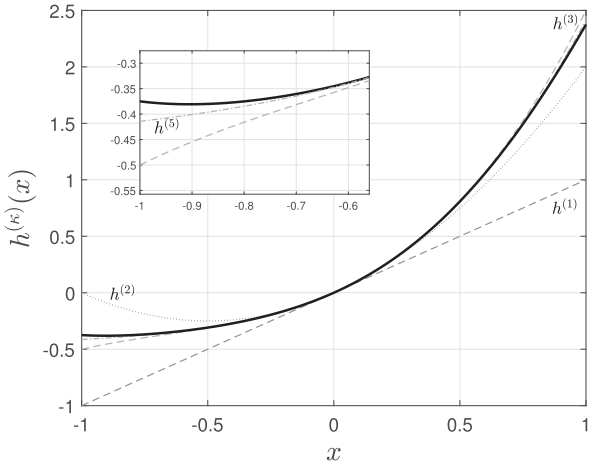


Fig. 1. Derivative of the value function $V^\star$ (solid black line) together with $h^{(\kappa)}$, for $\kappa = 1, 2, 3, 5$ for Example 1.

which satisfy Assumption 2 since $\mathrm{spec}(\bar{A}) \subset \mathbb{C}^-$ and which yields the positive definite solution $P_+ = I$ of the ARE (4). Setting $h : \mathbb{R}^2 \to \mathbb{R}^2$ as $h(x) = [h_1(x) \quad h_2(x)]^\top$, the corresponding quasi-linear PDE (44) is given by

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \frac{\partial h}{\partial x}(x) \begin{bmatrix} x_2 \\ -x_1 - \mu(1-x_1^2)x_2 - x_1^2 h_2(x) \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \end{bmatrix} + \begin{bmatrix} (x_1 x_2 - 1 - x_1 h_2(x))h_2(x) \\ h_1(x) + \frac{1}{2}(x_1^2 - 1)h_2(x) \end{bmatrix}. \quad (56)$$

By imposing the boundary condition (45), together with $h(0) = 0$, implies that the mapping h must exhibit the structure $h(x) = P_+ x + o(x^2) = x + o(x^2)$. As a matter of fact, by plugging the latter structure into (56), it is easy to verify that the higher order terms must be equal to zero and that $h(x) = x$ solves (44), (45). △

C. Nonlinear Optimal Control via Invariant Distribution

The main purpose of this section is to replicate, in the nonlinear setting, the spirit of the constructions discussed in Section II-C. More precisely, the latter suggests how the optimal solution of the LQR problem can be computed by performing a triangularizing change of coordinates on the underlying (linear) Hamiltonian dynamics: this is achieved, therein, by defining a linear coordinate transformation that involves the solution of the ARE (4).

The rest of this section is devoted to illustrating how the solution to an infinite-horizon optimal control problem may be indeed constructed on the basis of a certain *invariant distribution*. Interestingly, similarly to (44) in the statement of Proposition 4, also this approach leads to linear PDEs, although of a slightly different nature as discussed below. To this end, and to provide concise statements of the following results, let the mappings $f_i^{\mathbf{H}} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$, $i = 1, 2$, be defined according to:

$$f_1^{\mathbf{H}}(x, \lambda) := f(x) - g(x)g(x)^\top \lambda$$

$$f_2^{\mathbf{H}}(x, \lambda) := -\frac{\partial q}{\partial x}(x)^\top - \frac{\partial f}{\partial x}(x)^\top \lambda + \sum_{i=1}^{m} \lambda^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \lambda \quad (57)$$

so that the Hamiltonian dynamics (41) are compactly described by $(\dot{x}, \dot{\lambda}) = (f_1^{\mathbf{H}}(x, \lambda), f_2^{\mathbf{H}}(x, \lambda)) =: f^{\mathbf{H}}(x, \lambda)$. Let $\chi = (x, \lambda) \in \mathbb{R}^{2n}$. Furthermore, let $\tau_i : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$, $i = 1, \ldots, n$, be a collection of n independent smooth vector fields that solve, for $i = 1, \ldots, n$, the system of *linear* PDEs

$$0 = \frac{\partial f^{\mathbf{H}}}{\partial \chi}(\chi)\tau_i(\chi) - \frac{\partial \tau_i}{\partial \chi}(\chi) f^{\mathbf{H}}(\chi) - \sum_{j=1}^{n} c_{i,j}(\chi)\tau_j(\chi) \quad (58)$$

for all (x, λ) in a neighborhood of the origin, together with suitable functions $c_{i,j} : \mathbb{R}^{2n} \to \mathbb{R}$, $i = 1, \ldots, n$, $j = 1, \ldots, n$.

The following statement provides a counterpart of the results in Proposition 4, which revolve around the characterization of a certain *invariant manifold*, by identifying an *invariant distribution* that is related to the solution of a nonlinear optimal control problem (as explained in Proposition 6 below).

Proposition 5: Consider the optimal control problem described by (32) and (33), and suppose that Assumptions 2–4 hold. Let the distribution $\mathcal{D} : \mathbb{R}^{2n} \to \mathbb{R}^{2n \times n}$ be defined as

$$\mathcal{D} = \mathrm{span}\{\tau_1, \ldots, \tau_n\}. \quad (59)$$

i) Suppose that τ_i, $i = 1, \ldots, n$ satisfy (58) for some c_i's. Then $\mathcal{D}$ is an invariant distribution for the Hamiltonian dynamics (41).

ii) Suppose further[2] that $\mathcal{D}$ is involutive and such that $\dim(\mathcal{D}(0)) = n$. Let the gradients of the functions $\psi_{2,j} : \mathbb{R}^{2n} \to \mathbb{R}$, $j = 1, \ldots, n$, span the annihilator of $\mathcal{D}$, i.e.,

$$\mathcal{D}^\perp = \mathrm{span}\left\{ \frac{\partial \psi_{2,1}}{\partial \chi}, \ldots, \frac{\partial \psi_{2,n}}{\partial \chi} \right\} \quad (60)$$

and let the mappings $\Psi_2 = [\psi_{2,1}, \ldots, \psi_{2,n}]^\top$ and $\Psi_1 : \mathbb{R}^{2n} \to \mathbb{R}^n$ be such that the Jacobian matrix of

$$\Psi(x, \lambda) = \begin{bmatrix} \Psi_1(x, \lambda) \\ \Psi_2(x, \lambda) \end{bmatrix} \quad (61)$$

is nonsingular at $(x, \lambda) = (0, 0)$. Then the Hamiltonian dynamics (41) in the coordinates defined by $z = (z_1, z_2) = \Psi(x, \lambda) = (\Psi_1(x, \lambda), \Psi_2(x, \lambda))$ becomes

$$\dot{z}_1 = \tilde{f}_1^{\mathbf{H}}(z_1, z_2)$$

$$\dot{z}_2 = \tilde{f}_2^{\mathbf{H}}(z_2) \quad (62)$$

with $\tilde{f}_i^{\mathbf{H}}$, $\tilde{f}_2^{\mathbf{H}}$ obtained by transforming (57) via Ψ.

[2]It is implicitly assumed that item (ii) subsumes all the conditions stated in item (i). A similar consideration is also valid for item (iii) with respect to the conditions in items (i) and (ii).

iii) Suppose further that $\operatorname{span}\mathcal{D}(0) = \operatorname{span}[I \;\; P_+]^\top$. Then $z_1 = 0$ is a locally exponentially stable equilibrium point for $\dot{z}_1 = \tilde{f}_1^{\mathbf{H}}(z_1, 0)$. ∘

Proof: The claims in items i)–iii) can be shown by borrowing standard arguments and results from, e.g., [18]. To begin with, recalling the definition of Lie bracket of two vector fields, see [18, Sect. 1.2], it follows that the sum of the first two terms of the right-hand side of (58) coincides with $[f^{\mathbf{H}}, \tau_i]$. Thus, the existence of the functions $c_{i,j}$ in (58) tantamounts to the inclusion $[f^{\mathbf{H}}, \mathcal{D}] \subset \mathcal{D}$, which implies item i), by definition of invariance of $\mathcal{D}$ with respect to $f^{\mathbf{H}}$. The construction in item ii) follows immediately from [18, Thm. 1.4.1] and [18, Lemma 1.6.1]. Finally, item iii) is obtained by the structure of the change of coordinates Ψ (61), which is such that the variable z_2 has the property that $\operatorname{span}\{dz_2\}$ is the codistribution orthogonal to the distribution $\mathcal{D}$. As a consequence, by the hypothesis on the definition of $\mathcal{D}(0)$, z_1 and z_2 are associated with the stable and unstable, respectively, eigenvalues of the linearization of the vector field $\tilde{f}^{\mathbf{H}}$ around the origin. □

The following statement shows how the structure of the dynamics (62) and of the mapping Ψ in (61) can be leveraged to compute the optimal process.

Proposition 6: Consider the optimal control problem described by (32) and (33), and suppose that Assumptions 2–4 hold. Suppose that the conditions in items (i)–(iii) of Proposition 5 hold. Then there exists a nonempty set $\Omega \subset \mathbb{R}^n$, containing the origin, such that the optimal process $x^\star(t)$, $t \in \mathbb{R}_{\geq 0}$, is governed by

$$\begin{aligned} \dot{x}^\star(t) &= f(x^\star(t)) - g(x^\star(t))g(x^\star(t))^\top \Psi_2^{-1}(x^\star(t), 0) \\ x^\star(0) &= x_0 \end{aligned} \tag{63}$$

for all $x_0 \in \Omega$, where the mapping $\Psi_2^{-1} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is a (partial) inverse of Ψ_2 with respect to λ, i.e., such that $\lambda = \Psi_2^{-1}(x, \Psi_2(x, \lambda))$ for all $\lambda \in \mathbb{R}^n$. Moreover, the optimal feedback is given by $u^\star(x) = -g(x)^\top \Psi_2^{-1}(x, 0)$. △

Proof: By relying on the conclusions of items i)–iii) of Proposition 5 it follows that, for fixed $x(0) = x_0 \in \mathbb{R}^n$, the asymptotically stable trajectory of the Hamiltonian dynamics (41) is immediately obtained in the z-coordinates [see (62)] by selecting $z_2(0) = 0$. The latter condition in turn implies that $\lambda(0) = \Psi_2^{-1}(x_0, 0)$. This choice, by the (triangular) structure of (62), ensures that $z_2 \equiv 0$ and hence the optimal costate $\lambda(t) = \Psi_2^{-1}(x^\star(t), 0)$ for all $t \in \mathbb{R}_{\geq 0}$, which yields (63). □

Remark 6: Conclusions similar to those discussed in Proposition 6 have been obtained in [19], although with a few notable differences. First, Sassano deal with finite-horizon nonlinear optimal control problems. Then, in [19], the annihilating codistribution is computed directly with respect to the Hamiltonian vector field $f^{\mathbf{H}}$ rather than to the distribution $\mathcal{D}$ as in Proposition 5. As a consequence, on one hand, in the former case the need to solve the PDEs (58) is entirely circumvented. On the other hand, the nonuniqueness arising from the resulting codistribution of dimension $2n - 1$ [rather than exactly n as for $\mathcal{D}^\perp$ in (60)] should be subsequently addressed by solving a parameterized two-point boundary value problem (see [19, eq. (10)] and [19, Corollary. 1]).

While also Tarantino et al. [20] suggest an iterative scheme to compute nonlinear optimal control laws on the basis of constant optimal costate, this property is not achieved therein by relying on the notion of codistribution or first integral. This behavior is instead obtained, at each step, by performing a time-varying, linear change of coordinates to the linearization of the system along the trajectory computed at the previous iteration. ▲

Remark 7: With the objective of specializing the conditions of Propositions 5 and 6 to the case of linear systems and quadratic cost functionals, suppose that $f^{\mathbf{H}}(x, \lambda) = H(x, \lambda)$, with H defined in (9). Let $\mathcal{D}$ be described by the constant distribution

$$\mathcal{D}_V = \operatorname{span} \begin{bmatrix} I \\ P_+ \end{bmatrix}. \tag{64}$$

Thus, it can be easily verified that $\mathcal{D}_V$ satisfies (58) together with (constant) $c_{i,j}$, $i = 1, \ldots, n$, $j = 1, \ldots, n$ defined as

$$\begin{bmatrix} c_{1,1} & \ldots & c_{1,n} \\ \vdots & \ddots & \vdots \\ c_{n,1} & \ldots & c_{n,n} \end{bmatrix} := A - BB^\top P_+ . \tag{65}$$

Therefore, the functions $\psi_{2,j}$ are then given by the scalar entries of the mapping $\Psi_2(x, \lambda) = -P_+ x + \lambda$, the differentials of which span the annihilating codistribution of $\mathcal{D}_V$. Finally, as expected, the costate variable is obtained via $\lambda = \Psi_2^{-1}(x, 0) = P_+ x$. ▲

We conclude with the key observation.

> The specific occurrence of the ARE of Section II-C generalizes in the nonlinear setting to a *system of n, linear partial differential equations in the unknown mappings* $\tau_i : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ *and* $c_{i,j} : \mathbb{R}^{2n} \to \mathbb{R}$.

Example 3: Consider again the scalar optimal control problem introduced in Example 1. The associated Hamiltonian vector field $f^{\mathbf{H}} : \mathbb{R}^2 \to \mathbb{R}^2$ is given by

$$f^{\mathbf{H}} = \begin{bmatrix} x^2 - \lambda \\ -x - 2\lambda x \end{bmatrix}. \tag{66}$$

Straightforward, although tedious, computations show that the 1-D distribution $\mathcal{D}$ spanned by the vector field $\tau_1 : \mathbb{R}^2 \to \mathbb{R}^2$

$$\tau_1(x, \lambda) = \begin{bmatrix} x - \lambda - 2 \\ -\lambda - 3x - 2 \end{bmatrix} \tag{67}$$

zeroes the terms of order zero and one of the PDE (58), together with the scalar function $c_{1,1} : \mathbb{R}^2 \to \mathbb{R}$

$$c_{1,1}(x, \lambda) = \frac{2\lambda - 2x - 4\lambda x + x^2 + 2}{\lambda - x + 2}. \tag{68}$$

Furthermore, the distribution $\mathcal{D}$ satisfies the tangency condition in item (iii) of Proposition 5, is nonsingular around the origin and (trivially) involutive, hence it is completely integrable. It

can be shown that the corresponding function $\Psi_2 : \mathbb{R}^2 \to \mathbb{R}^2$ is given by

$$\Psi_2(x,\lambda) = \lambda - x - \frac{1}{2}\lambda x + \frac{1}{4}\lambda^2 - \frac{3}{4}x^2 . \tag{69}$$

Thus, according to the conclusions of Proposition 6, the (inverse) function

$$\Psi_2^{-1}(x,0) = x + 2\sqrt{x^2+1} - 2 \tag{70}$$

with $\Psi_2^{-1}(0,0) = 0$, provides an approximation of the gradient of $V^\star$ in (51), i.e., $dV^\star/dx = x^2 + x\sqrt{x^2+1}$. It is interesting to observe that, albeit the above approximation is obtained by solving (58) up to terms of order one, the expression in (70) is different from the second-order approximation (53) derived in Example 1, namely, $h^{(2)}(x) = x + x^2$, but they coincide up to order 2. △

D. Nonlinear Optimal Control Beyond HJB

To summarize the conclusions derived from the discussions in the previous sections, (38), (44), and (58) *all* reduce to the ARE (4) in the linear case. To put the overall design strategies into the correct perspective, note that in the case of (58), to complete the constructions of Proposition 6 one should also compute the annihilator of $\mathcal{D}$. This task tantamounts to solving a further system of n linear PDEs to determine the functions $\psi_{2,j}$, namely

$$\frac{\partial \psi}{\partial \chi}\tau_i(\chi) = 0, \quad i = 1,\dots,n. \tag{71}$$

Therefore, in the nonlinear setting, on the other hand, the ARE (4) can be equivalently generalized to *(i)* a single, quadratic PDE with positivity constraints (see (38) in Section IV-A); *(ii)* a system of n quasi-linear PDEs (see (44) in Section IV-B); *(iii)* a system of $2n$ linear PDEs (see (58) and (71) in Section IV-C), respectively.

V. Notion of Duality in Nonlinear Optimal Control

The discussions in the previous sections have highlighted the property that optimal control laws can be equivalently constructed by identifying certain invariant manifolds and distributions. These play the role of isolating convergent trajectories of the Hamiltonian dynamics $\Sigma_{\mathbf{H}}$ in such a way that the optimal process $x^\star$ can be computed. The strategy introduced in Section IV-B further lends itself to suggesting two reasonable ways of introducing the notion of duality for nonlinear optimal control problems. More precisely, this may be interpreted as either the construction of invariant manifolds that directly characterize the optimal costate variable, in place of the state, or of invariant *one-forms* (or covector fields), rather than manifolds, respectively. In fact, it has been shown in Section III that the former design philosophy yields a dual ARE, whereas, for the latter case, one-forms can be interpreted as the dual object with respect to mappings (see [18, Appendix A.6]). This objective is pursued in the two following sections.

A. Invariant Manifolds for Optimal Costate

Inspired by the arguments employed in Sections II-B and III within the linear setting, and hence with the aim of defining a manifold invariant with respect to the Hamiltonian dynamics (22) that permits the characterization of the optimal costate evolution, define

$$\mathcal{M}_d := \{(x,\lambda) \in \mathbb{R}^n \times \mathbb{R}^n : x - m(\lambda) = 0\} \tag{72}$$

with the mapping $m : \mathbb{R}^n \to \mathbb{R}^n$ to be defined, as discussed in the following statement.

Proposition 7: Consider the optimal control problem described by (32) and (33), and suppose that Assumptions 2–4 hold with the pair $(\bar{A}, \bar{Q})$ observable. Suppose that the mapping $m : \mathbb{R}^n \to \mathbb{R}^n$, $m \in \mathcal{C}^1$, $m(0) = 0$, satisfies the quasi-linear PDE

$$\begin{aligned} 0 = {} & f(m(\lambda)) - g(m(\lambda))g(m(\lambda))^\top\lambda + \frac{\partial m}{\partial \lambda}\frac{\partial q}{\partial x}(m(\lambda))^\top \\ & + \frac{\partial m}{\partial \lambda}(\lambda)\left(\frac{\partial f}{\partial x}(m(\lambda))^\top\lambda - \sum_{i=1}^m \lambda g_i(m(\lambda))\frac{\partial g_i}{\partial x}(m(\lambda))^\top\lambda\right) \end{aligned} \tag{73}$$

for all $\lambda \in \mathbb{R}^n$, together with the boundary condition

$$\frac{\partial m}{\partial \lambda}(0) = P_+^{-1} . \tag{74}$$

Then there exists a nonempty set $\Omega \subseteq \mathbb{R}^n \times \mathbb{R}^n$, containing the origin, such that the optimal costate variable $\lambda^\star(t)$, $t \in \mathbb{R}_{\geq 0}$, is governed by

$$\begin{aligned} \dot{\lambda} = {} & -\frac{\partial q}{\partial x}(m(\lambda))^\top - \frac{\partial f}{\partial x}(m(\lambda))^\top\lambda \\ & + \sum_{i=1}^m \lambda g_i(m(\lambda))\frac{\partial g_i}{\partial x}(m(\lambda))^\top\lambda \end{aligned} \tag{75a}$$

$$\lambda(0) = m^{-1}(x_0) \tag{75b}$$

for all $x_0 \in \Omega$. Moreover, the optimal process is given by

$$x^\star(t) = m(\lambda(t)). \tag{76}$$

∘

Proof: The claim is shown by relying on arguments similar to those employed in the proof of Proposition 4. Suppose that the mapping m satisfies the quasi-linear partial differential equation (73). Consider the variable $\xi = x - m(\lambda)$, suggested by the structure of $\mathcal{M}_d$ in (72). By virtue of (73), the dynamics of the variable ξ is then described by

$$\begin{aligned} \dot{\xi} = {} & f(m(\lambda)+\xi) - f(m(\lambda)) + \frac{\partial m}{\partial \lambda}(\lambda)\left(\frac{\partial q}{\partial x}(m(\lambda)+\xi)\right. \\ & \left. - \frac{\partial q}{\partial x}(m(\lambda)) + \frac{\partial f}{\partial x}(m(\lambda)+\xi)^\top\lambda - \frac{\partial f}{\partial x}(m(\lambda))^\top\lambda\right) \\ & - \frac{\partial m}{\partial \lambda}(\lambda)\sum_{i=1}^m \lambda^\top g_i(\xi + m(\lambda))\frac{\partial g_i}{\partial x}(\xi + m(\lambda))\lambda \end{aligned}$$

$$+\frac{\partial m}{\partial \lambda}(\lambda)\sum_{i=1}^{m}\lambda^\top g_i(m(\lambda))\frac{\partial g_i}{\partial x}(m(\lambda))\lambda$$

$$=:\ \delta_d(\xi,\lambda) \tag{77}$$

with δ_d such that $\delta_d(0,\lambda)\equiv 0$. The boundary condition (74) ensures that m is a local diffeomorphism around the origin, since its Jacobian matrix is nonsingular, where positive definiteness of the underlying P_+ is guaranteed by observability of the pair $(\bar{A},\bar{Q})$. Therefore there exists a nonempty set Ω, containing the origin, such that the inverse mapping m^{-1} is well defined. The selection of the initial condition for the costate variable λ in (75) is

$$\lambda(0)=m^{-1}(x_0) \tag{78}$$

which ensures that $0=x^\star(0)-m(\lambda(0))=x_0-m(m^{-1}(x_0))$. The latter in turn implies that $\xi(0)=0$ and hence $\xi(t)=0$, for all $t\in\mathbb{R}_{\geq 0}$. Furthermore, the boundary condition (74) is such that the manifold defined by $\{(\lambda,x)\in\mathbb{R}^n\times\mathbb{R}^n : 0=\lambda-m^{-1}(x)\}$ is tangent to the stable subspace of the linearization of (41) at the origin, which is a well-defined n-dimensional subspace by virtue of Assumption 2, hence locally coincides with the stable submanifold of (41). □

By relying on arguments identical to those employed in the proof of Proposition 2, provided the columns of $g(m(\lambda(t)))$ are linearly independent for all $t\in\mathbb{R}$, the optimal control law can be computed according to

$$u^\star(t)=-g(m(\lambda(t)))^\dagger\left(f(m(\lambda(t)))-\frac{\partial m}{\partial\lambda}(\lambda(t))\dot{\lambda}(t)\right) \tag{79}$$

where $\lambda:\mathbb{R}_{\geq 0}\to\mathbb{R}^n$ is obtained via (75). In fact, by the structure of $\mathcal{M}_d$ and the fact that $\xi\equiv 0$, one has that the optimal process, which satisfies $\dot{x}^\star(t)=f(x^\star(t))+g(x^\star(t))u^\star(t)$, is also such that $x^\star(t)=m(\lambda(t))$ for all $t\in\mathbb{R}_{\geq 0}$. Therefore, the following implication can be established

$$\begin{aligned}&\dot{x}^\star(t)-f(x^\star(t))\in\operatorname{Im}g(x^\star(t))\\ &\Rightarrow\ \frac{\partial m}{\partial\lambda}\dot{\lambda}(t)-f(m(\lambda(t)))\in\operatorname{Im}g(m(\lambda(t)))\end{aligned} \tag{80}$$

for all $t\geq 0$. Therefore, define the error variable $\vartheta=x-m(\lambda)$, with x denoting the state of (32), and observe that, by construction, $\vartheta(0)=0$. Finally, the choice of u in (32) according to (79) ensures that $\dot{\vartheta}|_{\vartheta=0}=0$ and hence that the state variable x of (32) evolves as $m(\lambda)$, and hence as $x^\star$, for all $t\in\mathbb{R}_{\geq 0}$.

B. On the Gap Between (44), (58), and (73)

Since several alternative perspectives on the characterization of optimal control laws for nonlinear systems have been provided, it is worth investigating the connections between the existence of solutions to such conditions. To begin with, by the requirements of Assumptions 2–4 one has that the nonlinear Hamiltonian dynamics $\Sigma_{\mathbf{H}}$ in (41) admits a well-defined, unique, stable invariant manifold [21]. Therefore, the quasi-linear partial differential equation (44), which characterizes such a manifold, indeed admits a solution. As a consequence, and by observing that the partial differential equation essentially characterizes the inverse mapping of the gradient of the value function, also the existence of the solution to (73) is guaranteed by the inverse function theorem. Finally, [22, Appendix 2] ensures the existence of the solution of the partial differential equation (58) by noting that the regularity properties of the stable invariant manifold imply the existence and uniqueness of a foliation associated with such a manifold. In summary, there is no gap between the conditions provided by (44), (58), and (73) in the characterization of optimal feedback control laws for nonlinear systems over an infinite horizon.

C. Numerical Case Studies

The results above are validated by means of two academic case studies.

1) Scalar Nonlinear System: Consider again the nonlinear optimal control problem introduced in Example 1. The quasi-linear partial differential equation (73) for the costate reduces to

$$0=\frac{dm}{d\lambda}(\lambda)(m(\lambda)+2m(\lambda)\lambda)+m(\lambda)^2-\lambda,\ \ \frac{dm}{d\lambda}(0)=1 \tag{81}$$

and $m(0)=0$. Note that, by introducing the function $p:\mathbb{R}\to\mathbb{R}$ defined as $p(\lambda):=m(\lambda)^2$ one has that

$$\frac{dp}{d\lambda}(\lambda)=2\frac{dm}{d\lambda}(\lambda)m(\lambda)$$

and hence (81) can be written as the linear ordinary differential equation

$$\frac{dp}{d\lambda}(\lambda)=-\frac{2}{1+2\lambda}p(\lambda)+\frac{2\lambda}{1+2\lambda}\,. \tag{82}$$

The latter can then be immediately solved to obtain the solution sought for (81), given by

$$m(\lambda)=\frac{\lambda}{\sqrt{1+2\lambda}} \tag{83}$$

which is well defined for $\lambda>-1/2$. Thus, one has that the optimal control law may be obtained via $u(t)=-\lambda(t)$, where λ is governed by the asymptotically stable nonlinear systems

$$\dot{\lambda}=-\lambda\sqrt{1+2\lambda} \tag{84}$$

initialized at $\lambda(0)=m^{-1}(x_0)$. Furthermore, as it is easy to verify, the inverse function $m^{-1}(x)$, with m defined in (83), coincides with the derivative of the value function $V^\star$ given in (51) and, moreover, it is such that $m^{-1}:\mathbb{R}\to\mathbb{R}_{>-\frac{1}{2}}$. Fig. 2 illustrates the comparison between the control law designed via the costate variable obtained by considering the original Hamiltonian dynamics (41) or on the basis of the costate trajectory computed by forward propagating (84). In particular, the solid black line denotes the optimal control law $u^\star(t)=-\lambda(t)$, where λ is obtained via (41) initialized by setting $x(0)=x_0$ and $\lambda(0)=(dV^\star/dx)(x_0)$. Such an optimal control law is then

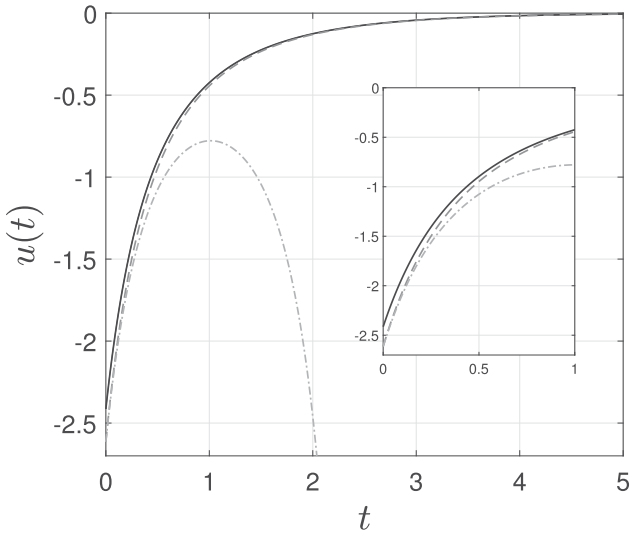


Fig. 2. Time history of the optimal control law $u^\star$ (solid black line) compared with the time histories of the control law obtained by integrating (84) initialized at $\lambda(0) = m^{-1}(x_0) + \varepsilon$ (gray dashed line) and (41) initialized at $\lambda(0) = (dV^\star/dx)(x_0) + \varepsilon$ (light gray, dash-dotted line), with $\varepsilon = 0.2$, for the example in Section V-C1.

compared with perturbed instances of the costate dynamics (41) and of (75). More precisely, the gray dashed line depicts the control law $u(t) = -\lambda(t)$ with the costate λ obtained by integrating (84) initialized at $\lambda(0) = m^{-1}(x_0) + \varepsilon$ and the gray dash-dotted line describes the control law obtained by considering the costate variable given by (41) initialized at $\lambda(0) = (dV^\star/dx)(x_0) + \varepsilon$, with $\varepsilon = 0.2$, thus showing the improved robustness properties of the former over the latter.

2) Planar Nonlinear System: Consider the planar nonlinear system described by (32) with

$$f(x) = \begin{bmatrix} -2x_1x_2 + 2x_2^3 \\ -x_1 + x_2^2 \end{bmatrix}, \quad g(x) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{85}$$

together with the cost functional (33), with

$$q(x) = \frac{1}{2}x_2^2 \tag{86}$$

and $R = 1$. The partial differential equation (73) admits a solution $m : \mathbb{R}^2 \to \mathbb{R}^2$ described by

$$m(\lambda) = \begin{bmatrix} \frac{4\sqrt{2}\lambda_1^3 - 5\lambda_1^2 + \sqrt{2}\lambda_1 + 2\lambda_2^2 + \lambda_2}{(2\sqrt{2}\lambda_1 - 1)^2} \\ -\frac{\lambda_1 + \sqrt{2}\lambda_2}{2\sqrt{2}\lambda_1 - 1} \end{bmatrix} \tag{87}$$

which verifies the boundary condition

$$\frac{\partial m}{\partial \lambda}\Big|_{\lambda=0} = \begin{bmatrix} \sqrt{2} & 1 \\ 1 & \sqrt{2} \end{bmatrix} = P_+^{-1}$$

where P_+ denotes the positive definite solution of the ARE (4) associated with the LQ approximation of (85) and (86). Therefore, by combining the conclusions of Propositions 2 and 7, it follows that (75) with m given by (87) consists of a (locally around the origin) asymptotically stable system that generates the trajectories of the optimal costate variable.

D. Nonlinear Optimal Control via Invariant One-Forms

The purpose of this section consists in the characterization of optimal control laws by means of an invariant one-form, see for instance [18, Appendix A.6] for further discussions about the definition and properties of one-forms. As it is relevant for the following derivations, it is worth recalling that a generic one-form $v : \mathbb{R}^n \to T^\star\,\mathbb{R}^n$ can be expressed in local coordinates according to

$$v(x)dx = \sum_{i=1}^{n} v_i(x)dx_i = \begin{bmatrix} v_1(x) & \dots & v_n(x) \end{bmatrix} dx \tag{88}$$

for some smooth functions $v_i : \mathbb{R}^n \to \mathbb{R}$, with $dx = [dx_1 \quad \dots \quad dx_n]^\top$. Furthermore, the derivative of a one-form along a generic vector field $f : \mathbb{R}^n \to \mathbb{R}^n$, denoted $L_f v(x)$, is given by

$$L_f v(x) = \left[\frac{\partial v^\top}{\partial x}(x) f(x)\right]^\top + v(x)\frac{\partial f}{\partial x}(x)\,. \tag{89}$$

By mimicking now the definition of the manifold $\mathcal{M}$ in (43), consider the one-forms $\omega_j : \mathbb{R}^{2n} \to \mathbb{R}^{1\times 2n}$ defined as

$$\begin{aligned} \omega_j(x,\lambda) &= p_{j,1}(x)dx_1 + \dots + p_{j,n}dx_n - d\lambda_j \\ &=: p_j(x)dx - e_j^\top d\lambda \end{aligned} \tag{90}$$

for $j = 1, \dots, n$, with the functions $p_{j,i} : \mathbb{R}^n \to \mathbb{R}$ to be determined and where $e_j \in \mathbb{R}^n$ denotes the jth element of the canonical basis. The codistribution spanned by the one-forms $\omega_1, \dots, \omega_n$ can be compactly denoted by stacking the one-forms on the rows of

$$\omega(x,\lambda) = \begin{bmatrix} \omega_1(x,\lambda) \\ \vdots \\ \omega_n(x,\lambda) \end{bmatrix} =: P(x)dx - d\lambda \tag{91}$$

where the rows of the matrix-valued function P consist of the row vectors p_j introduced in (90). Suppose further that the (partial) one-forms p_j are exact, namely that there exist functions $\phi_j : \mathbb{R}^n \to \mathbb{R}$ with the property that $d\phi_j(x) = p_j(x)$. As a consequence, the one-forms ω_j are exact and they can be described by the gradients of the functions $\phi_j(x) - \lambda_j$. Therefore, invariance of the one-forms ω along the vector field $f^{\mathbf{H}}$ in (57) is enforced provided one has that

$$\dot{\omega}_j = L_{f^{\mathbf{H}}}\omega_j\Big|_{\omega=0} = 0 \tag{92}$$

for $j = 1, \dots, n$. Note that, by the structure of ω in (91) and the integrability condition on p_j, it follows that $\lambda = \phi(x)$, with $\phi(x) = [\phi_1(x) \quad \dots \quad \phi_n(x)]^\top$, whenever $\omega(x,\lambda) = 0$. The above constructions is formalized in the following statement, which characterizes the optimal solution in terms of invariant codistributions.

Proposition 8: Consider the optimal control problem described by (32) and (33), and suppose that Assumptions 2–4

hold. Suppose that the exact one-forms p_j, $j = 1, \ldots, n$ satisfy the second-order PDE (93) (at the bottom of this page) for all $x \in \mathbb{R}^n$, with $P(x) = [p_1(x)^\top \quad \ldots \quad p_n(x)^\top]^\top$, together with the boundary condition $P(0) = P_+$. Then there exists a nonempty set $\Omega \subset \mathbb{R}^n$, containing the origin, such that the optimal process $x^\star(t)$, $t \in \mathbb{R}_{\geq 0}$, is governed by

$$\begin{aligned} \dot{x}^\star(t) &= f(x^\star(t)) - g(x^\star(t))g(x^\star(t))^\top \phi(x^\star(t)) \\ x^\star(0) &= x_0 \end{aligned} \tag{94}$$

for all $x_0 \in \Omega$, hence the optimal feedback is given by $u^\star(x) = -g(x)^\top \phi(x)$. ∘

Proof: The partial differential equation (93) is obtained by imposing individual invariance of the one-forms ω_j along the vector field $f^{\mathbf{H}}$ and then stacking the equations together. More precisely, one has that

$$\left(\frac{\partial \omega_j^\top}{\partial x} f^{\mathbf{H}} \right)^\top = (f(x) - g(x)g(x)^\top \lambda)^\top \frac{\partial p_j}{\partial x} dx \tag{95}$$

and

$$\begin{aligned} \omega_j \frac{\partial f^{\mathbf{H}}}{\partial (x, \lambda)} &= p_j(x) \left(\frac{\partial f}{\partial x}(x) - \frac{\partial}{\partial x} \left(g(x)g(x)^\top \lambda \right) \right) dx \\ &+ e_j^\top \frac{\partial^2 q}{\partial x^2} dx - e_j^\top \frac{\partial}{\partial x} \left(\sum_{i=1}^{\top} \lambda^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \lambda \right) dx \\ &+ e_j^\top \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x}(x)^\top \lambda \right) dx - p_j(x)g(x)g(x)^\top d\lambda \\ &+ e_j^\top \frac{\partial f}{\partial x}(x)^\top d\lambda + 2e_j^\top \sum_{i=1}^{m} \left(\frac{\partial g_i}{\partial x}(x) g_i(x)^\top \lambda \right) d\lambda \,. \end{aligned} \tag{96}$$

Therefore, the partial differential equation (93) is obtained by stacking the individual equations $L_{f^{\mathbf{H}}} \omega_j|_{\omega=0} = 0$ into rows and recalling that $\omega(x, \lambda) = 0$ implies that $d\lambda = P(x)dx$ and $\lambda = \phi(x)$. □

By recalling the connection between ϕ and P in (93), namely that P contains first-order derivatives of the functions in ϕ, it follows that the latter equation is, in fact, a second-order partial differential equation in the unknown ϕ. A few simplifying assumptions are considered for the PDE (93) in the following remark.

Remark 8: Suppose that the input vector field in (32) is constant, namely $g(x) \equiv B$. Then the PDE (93) becomes

$$\begin{aligned} 0 = &\begin{bmatrix} (f(x) - BB^\top \phi(x))^\top \frac{\partial p_1}{\partial x} \\ \vdots \\ (f(x) - BB^\top \phi(x))^\top \frac{\partial p_n}{\partial x} \end{bmatrix} + \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x}(x)^\top \lambda \right) \Big|_{\lambda = \phi(x)} \\ &+ \frac{\partial^2 q}{\partial x^2}(x) + P(x) \frac{\partial f}{\partial x}(x) - P(x)BB^\top P(x) + \frac{\partial f}{\partial x}(x)^\top P(x) \,. \end{aligned} \tag{97}$$

If it is further assumed that the underlying system is in fact linear, namely $f(x) = Ax$, and that $q(x) = (1/2)x^\top Qx$, then it is interesting to observe that (93) reduces, *again*, to the classic ARE (4), if invariance of the one-form $Pdx - d\lambda$, with P constant matrix, is enforced. ▲

Example 4: Consider again the nonlinear optimal control problem introduced in Example 1, which has been employed to illustrate the constructions in the previous sections. Since $g(x) = B = 1$, one must consider the simplified partial differential equation (97), which reduces to

$$0 = \frac{\partial^2 \phi}{\partial x^2}(x^2 - \phi(x)) + 2\phi(x) + 1 + 4x \frac{\partial \phi}{\partial x} - \left(\frac{\partial \phi}{\partial x} \right)^2 \tag{98}$$

in the unknown function $\phi : \mathbb{R} \to \mathbb{R}$. Let now

$$\phi(x) = x + \frac{a}{2}x^2 + \frac{b}{3}x^3 + \frac{c}{4}x^4 + \frac{d}{5}x^5 \tag{99}$$

where the coefficient of the linear term is selected to satisfy the boundary condition $P(0) = (\partial \phi / \partial x)_{x=0} = P_+ = 1$, such that $\partial \phi / \partial x = 1 + ax + bx^2 + cx^3 + dx^4$ and $\partial^2 \phi / \partial x^2 = a + 2bx + 3cx^2 + 4\,dx^3$. Replacing the above expressions into (98) and truncating to the terms of order four, one obtains

$$\begin{aligned} 0 = (6 - 3a)x + \left(-\frac{3}{2}a^2 + 6a - 4b \right) x^2 \\ + \left(\frac{20}{3}b - 5c - \frac{10}{3}ab \right) x^3 + \left(\frac{15}{2}c - \frac{5}{3}b^2 - 6d - \frac{15}{4}ac \right) x^4 \end{aligned} \tag{100}$$

which, solved with respect to the coefficients $a \in \mathbb{R}, b \in \mathbb{R}, c \in \mathbb{R}, d \in \mathbb{R}$, yields the unique solution $a = 2$, $b = 3/2$, $c = 0$ and $d = -5/8$, thus recovering the solution $h^{(5)}$ constructed in (53). △

$$\begin{aligned} 0 = &\begin{bmatrix} (f(x) - g(x)g(x)^\top(x)\phi(x))^\top \frac{\partial p_1}{\partial x} \\ \vdots \\ (f(x) - g(x)g(x)^\top \phi(x))^\top \frac{\partial p_n}{\partial x} \end{bmatrix} + \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x}(x)^\top \lambda \right) \Big|_{\lambda = \phi(x)} + \frac{\partial^2 q}{\partial x^2}(x) - \sum_{i=1}^{m} \phi(x)^\top g_i(x) \frac{\partial g_i}{\partial x}(x)^\top \phi(x) \\ &+ P(x) \left(\frac{\partial f}{\partial x}(x) - \frac{\partial}{\partial x}(g(x)g(x)^\top \lambda) \right) \Big|_{\lambda = \phi(x)} - P(x)g(x)g(x)^\top P(x) + \frac{\partial f}{\partial x}(x)^\top P(x) + 2 \sum_{i=1}^{m} \frac{\partial g_i}{\partial x}(x) g_i(x)^\top \phi(x) \end{aligned} \tag{93}$$

VI. Conclusion

Within the framework of optimal control problems defined over an infinite-horizon, the first objective of this work has been to revisit the LQR problem with the aim of clarifying the structural motivations underlying the ubiquitous appearance of the ARE, and of leveraging these insights to propose novel perspectives and computational tools, both in the linear and nonlinear settings. It has been shown that, while all strategies lead to the ARE, the mechanisms leading to it differ substantially depending on whether one adopts dynamic programming, related to the construction of a value function, or Pontryagin's minimum principle, related to the existence of certain invariant hyperplanes. A further interpretation has been proposed, showing that the ARE is intimately involved in the construction of a coordinates transformation that render the Hamiltonian system block triangular. This perspective naturally leads to a dual Riccati equation and suggests a numerically robust method to compute the optimal costate via an asymptotically stable linear system.

The conceptual understanding gained from the linear case has then been exploited to address nonlinear optimal control problems. While the three strategies considered, namely the characterization of the value function, of a suitable invariant hyperplane, and of a triangularizing coordinate transformations, reduce to the ARE in the linear setting, they give rise to fundamentally different mathematical objects in the nonlinear case. This distinction allows one to move beyond the classical Hamilton–Jacobi–Bellman framework and to enrich the set of available computational tools. In particular, it has been shown that the optimal feedback can be equivalently obtained either by characterizing suitable invariant manifolds or by identifying invariant distributions of the Hamiltonian dynamics. Unlike the linear case, these objects are no longer identical and lead to significantly different formulations: the former results in a system of n quasi-linear partial differential equations, whereas the latter yields a system of $2n$ linear PDE, both alternatives to the classic scalar quadratic HJB PDE. Finally, the notion of duality introduced in the linear setting has been shown to admit two natural, and somewhat different, extensions in the nonlinear framework.

The derivation of conditions that are equivalent to the celebrated HJB partial differential equation although profoundly different in nature may pave the way for advancements in the several contexts in which the HJB equation indeed plays a crucial role. To mention a couple of examples, it is worth recalling that the wide-spread approaches based on reinforcement learning (RL) or adaptive dynamic programming (ADP) revolve around the use of the Bellman and Hamilton–Jacobi–Bellman equations: replacing the underlying condition in such frameworks could lead to a paradigm shift opening research directions that have been unforseen hitherto.

Furthermore, even from a computational perspective, it would be interesting to explore whether linearity of the newly established equations could yield alternative, numerically reliable or more efficient, algorithms to construct optimal control laws for nonlinear systems. The construction of the optimal control law via an invariant distribution appears promising in this respect and may deserve further attention. In fact, although a series expansion approach could be equivalently applied to all the derived equations (including those already available in the literature), the fact that (63) is based on an inverse mapping may further enhance the approximation capabilities of the approach for a fixed truncation of the series expansion (see the concluding remarks of Example 3).

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